

## Td Limites ln

**Exercice 1** Déterminer les limites suivantes :

1  $\lim_{x \rightarrow +\infty} \ln(x+2)$

2  $\lim_{x \rightarrow 0} (\ln x)^2$

3  $\lim_{x \rightarrow 0} \frac{1}{\ln x}$

4  $\lim_{x \rightarrow +\infty} \ln(x^2 + x + 1)$

5  $\lim_{x \rightarrow -\infty} \ln(x^2 + x + 1)$

6  $\lim_{x \rightarrow +\infty} \ln \left[ \frac{x+1}{x^2+x+1} \right]$

7  $\lim_{x \rightarrow 1^+} \ln \left[ \frac{x+2}{x-1} \right]$

8  $\lim_{x \rightarrow +\infty} \frac{\ln(x+2)}{\ln(x+1)}$

9  $\lim_{x \rightarrow 0^+} \frac{\ln x + 2}{\ln x + 1}$

10  $\lim_{x \rightarrow +\infty} \frac{\ln x}{\sqrt{x}}$

11  $\lim_{x \rightarrow +\infty} \frac{\ln x}{x^2}$

12  $\lim_{x \rightarrow +\infty} (\ln x - \sqrt{x})$

13  $\lim_{x \rightarrow 0^+} x^2 \ln x$

14  $\lim_{x \rightarrow 0^+} \sqrt{x} \ln x$

15  $\lim_{x \rightarrow +\infty} (\ln x - 2x)$

16  $\lim_{x \rightarrow +\infty} \frac{\ln(x^2+1)}{\ln(x+1)}$

17  $\lim_{x \rightarrow 0} \frac{\ln(1+2x)}{x}$

18  $\lim_{x \rightarrow 0} \frac{\ln(1+x^2)}{x}$

19  $\lim_{x \rightarrow -\infty} \frac{\ln(1+e^x)}{e^x}$

20  $\lim_{x \rightarrow 0} e^x \ln x$

21  $\lim_{x \rightarrow -\infty} \frac{\ln(1+e^x)}{x}$

22  $\lim_{x \rightarrow +\infty} \frac{\ln(1+e^x)}{x}$

23  $\lim_{x \rightarrow 0} \frac{\ln(1+x^2)}{3x}$

24  $\lim_{x \rightarrow 0} \frac{\ln(e^x - 1)}{x}$

25  $\lim_{x \rightarrow 0^+} \frac{2 \ln^2 x}{x}$

26  $\lim_{x \rightarrow 0} \frac{2x+1}{x} \ln x$

27  $\lim_{x \rightarrow +\infty} \frac{2x+1}{x} \ln x$

28  $\lim_{x \rightarrow 0} \frac{\ln x - 1}{2x}$

29  $\lim_{x \rightarrow +\infty} \frac{\ln x - 1}{2x}$

30  $\lim_{x \rightarrow 2^+} \ln(2-x)$

31  $\lim_{x \rightarrow +\infty} \ln(x^2 - 2x)$

32  $\lim_{x \rightarrow -\infty} \ln(x^2 + 4x)$

33  $\lim_{x \rightarrow 0} \ln \left( \frac{2x-1}{x-2} \right)$

34  $\lim_{x \rightarrow 0^+} \ln \left( \frac{1+x}{x} \right)$

35  $\lim_{x \rightarrow +\infty} \ln \left( \frac{1-x}{-x-1} \right)$

36  $\lim_{x \rightarrow -\infty} \ln \left( \frac{x^3+x}{-2x^2-1} \right)$

$$37 \quad \lim_{x \rightarrow 0^+} \ln|x^2 + x|$$

$$38 \quad \lim_{x \rightarrow 0^-} \ln|x^2 + x|$$

$$39 \quad \lim_{x \rightarrow 1} \ln|x^2 - 1|$$

$$40 \quad \lim_{x \rightarrow +\infty} \ln|x^2 - 1|$$

$$41 \quad \lim_{x \rightarrow +\infty} \frac{2x - 1}{x \ln x - x}$$

$$42 \quad \lim_{x \rightarrow 0^+} \frac{2x - 1}{x \ln x + x}$$

$$43 \quad \lim_{x \rightarrow +\infty} \frac{-1}{x^2 \ln x}$$

$$44 \quad \lim_{x \rightarrow 0^+} \frac{2 \ln x - 3}{3 \ln x + 1}$$

$$45 \quad \lim_{x \rightarrow +\infty} \frac{2 \ln x - 3}{3 \ln x + 1}$$

$$46 \quad \lim_{x \rightarrow 0^+} \frac{\ln x - 3x}{2 \ln x + x}$$

$$47 \quad \lim_{x \rightarrow +\infty} \frac{\ln x - 3x}{2 \ln x + x}$$

$$48 \quad \lim_{x \rightarrow 0^+} \frac{-1}{x^2 \ln x}$$

$$49 \quad \lim_{x \rightarrow +\infty} x - \ln x$$

$$50 \quad \lim_{x \rightarrow +\infty} \frac{1 - \ln x}{x}$$

$$51 \quad \lim_{x \rightarrow 0^+} \frac{1 - \ln x}{x}$$

$$52 \quad \lim_{x \rightarrow +\infty} x \ln x - x$$

$$53 \quad \lim_{x \rightarrow 0} x \ln x - \ln x$$

$$54 \quad \lim_{x \rightarrow 0^+} x^n \ln^2 x$$

$$55 \quad \lim_{x \rightarrow 0^+} x(\ln x - 1)$$

$$56 \quad \lim_{x \rightarrow +\infty} \frac{\ln(x^2)}{x}$$

$$57 \quad \lim_{x \rightarrow +\infty} \frac{(x - 1) \ln x}{x}$$

$$58 \quad \lim_{x \rightarrow 0^+} \frac{(x - 1) \ln x}{x}$$

$$59 \quad \lim_{x \rightarrow +\infty} x \ln \left( \frac{x + 1}{x - 1} \right)$$

$$60 \quad \lim_{x \rightarrow +\infty} x \ln \left( \frac{2x + 1}{2x + 3} \right)$$

$$61 \quad \lim_{x \rightarrow -\infty} \frac{\ln(x^2)}{x}$$

$$62 \quad \lim_{x \rightarrow 0^+} \left( \ln x + \frac{1}{\ln x} \right)$$

$$63 \quad \lim_{x \rightarrow +\infty} \frac{x^2}{x - 1} - \ln(x)$$

$$64 \quad \lim_{x \rightarrow +\infty} \frac{-1}{x^2 \ln x}$$

$$65 \quad \lim_{x \rightarrow 0^+} x \ln^2 x$$

$$66 \quad \lim_{x \rightarrow 0^+} \sqrt{x} \ln \left( \frac{x + 1}{x} \right)$$

$$67 \quad \lim_{x \rightarrow 0^+} x^3 \ln x$$

$$68 \quad \lim_{x \rightarrow 0} \frac{\ln x - \ln 2}{x - 2}$$

$$69 \quad \lim_{x \rightarrow 0} \frac{\ln(a + x) - \ln a}{x}$$

$$70 \quad \lim_{x \rightarrow 0} \frac{\ln x - 2}{\ln x + 1}$$

$$71 \quad \lim_{x \rightarrow 0^+} \tan x \ln x$$

$$72 \quad \lim_{x \rightarrow \frac{\pi}{4}} \frac{\ln(\tan x)}{\sin x - \cos x}$$

$$73 \quad \lim_{x \rightarrow 0^+} \frac{\ln(\cos x)}{x^2}$$

$$74 \quad \lim_{x \rightarrow +\infty} \frac{x \ln x}{x + 1}$$

$$75 \quad \lim_{x \rightarrow +\infty} x \ln \left( 1 + \frac{1}{x} \right)$$

$$76 \quad \lim_{x \rightarrow +\infty} \frac{\ln(1 + x)}{1 + x^2}$$

$$77 \quad \lim_{x \rightarrow +\infty} \frac{x \ln x}{x + 1}$$

$$78 \quad \lim_{x \rightarrow 0^+} \frac{\ln x}{1 - \ln x}$$

$$79 \quad \lim_{x \rightarrow +\infty} \frac{\ln x}{1 - \ln x}$$

$$80 \quad \lim_{x \rightarrow 0^+} \frac{1}{x^2} \ln \left( \frac{1 - x^2}{\cos x} \right)$$

$$81 \quad \lim_{x \rightarrow 1} -x + \ln \left| \frac{x}{x-1} \right|$$

$$82 \quad \lim_{x \rightarrow 1} -x + \ln \left| \frac{x}{x-1} \right|$$

$$83 \quad \lim_{x \rightarrow -\infty} -x + \ln \left| \frac{x}{x-1} \right|$$

$$84 \quad \lim_{x \rightarrow +\infty} -x + \ln \left| \frac{x}{x-1} \right|$$

$$85 \quad \lim_{x \rightarrow +\infty} x \ln \left| \frac{x+1}{x-1} \right|$$

$$86 \quad \lim_{x \rightarrow -\infty} \frac{\ln \sqrt{x^2 - 1}}{x^2 - 1}$$

$$87 \quad \lim_{x \rightarrow 1} \frac{\ln(x^2 - 2x + 2)}{(x-1)^2}$$

$$88 \quad \lim_{x \rightarrow 0} \frac{\ln(1 + \sqrt{x})}{1 - \sqrt{x+1}}$$

$$89 \quad \lim_{x \rightarrow +\infty} \left( \ln x + \frac{1}{\ln x} \right)$$

$$90 \quad \lim_{x \rightarrow 0^+} \frac{\ln x}{1 + x^2}$$

$$91 \quad \lim_{x \rightarrow +\infty} \frac{\ln x}{1 + x^2}$$

$$92 \quad \lim_{x \rightarrow +\infty} \frac{x^3}{\ln x}$$

$$93 \quad \lim_{x \rightarrow +\infty} \frac{\ln(1 + \ln x)}{x}$$

$$94 \quad \lim_{x \rightarrow +\infty} x \ln x - \ln x$$

$$95 \quad \lim_{x \rightarrow +\infty} \sqrt{x} \ln(x)$$

$$96 \quad \lim_{x \rightarrow 0} \frac{x}{\ln(1 + \sin x)}$$

$$97 \quad \lim_{x \rightarrow 0} \frac{\ln(1 + 2x)}{\tan x}$$

$$98 \quad \lim_{x \rightarrow +\infty} \frac{\ln x}{x-1}$$

$$99 \quad \lim_{x \rightarrow 1} \frac{\ln x}{x-1}$$

**Exercice 2** Déterminer les limites suivantes :

$$1 \quad \lim_{x \rightarrow +\infty} e^x$$

$$2 \quad \lim_{x \rightarrow -\infty} e^x$$

$$3 \quad \lim_{x \rightarrow +\infty} \frac{e^x}{x}$$

$$4 \quad \lim_{x \rightarrow +\infty} \frac{e^x}{x^3}$$

$$5 \quad \lim_{x \rightarrow +\infty} x e^{-x}$$

$$6 \quad \lim_{x \rightarrow +\infty} x^2 e^{-x}$$

$$7 \quad \lim_{x \rightarrow +\infty} \frac{e^{2x}}{e^x + 1}$$

$$8 \quad \lim_{x \rightarrow +\infty} \frac{e^{3x} + e^x}{e^{3x} - 2}$$

$$9 \quad \lim_{x \rightarrow -\infty} e^{2x}$$

$$10 \quad \lim_{x \rightarrow -\infty} \frac{e^x}{1 + e^x}$$

$$11 \quad \lim_{x \rightarrow +\infty} \frac{e^{x+2}}{e^x}$$

$$12 \quad \lim_{x \rightarrow +\infty} \frac{e^{2x}}{e^{x-1}}$$

$$13 \quad \lim_{x \rightarrow +\infty} (e^{x+1} - e^x)$$

$$14 \quad \lim_{x \rightarrow +\infty} (e^{2x} - e^x)$$

$$15 \quad \lim_{x \rightarrow 0} \frac{e^x - 1}{x}$$

$$16 \quad \lim_{x \rightarrow 0} \frac{e^{2x} - 1}{x}$$

$$17 \quad \lim_{x \rightarrow 0} \frac{e^x - e^{-x}}{x}$$

$$18 \quad \lim_{x \rightarrow 0} \frac{e^x - 1 - x}{x^2}$$

$$19 \quad \lim_{x \rightarrow 0} \frac{e^{3x} - 1 - 3x}{x^2}$$

$$20 \quad \lim_{x \rightarrow +\infty} \frac{x + e^x}{e^x}$$

$$21 \quad \lim_{x \rightarrow +\infty} \frac{e^x - x}{e^x}$$

$$22 \quad \lim_{x \rightarrow -\infty} (x e^x)$$

$$23 \quad \lim_{x \rightarrow -\infty} \frac{x}{e^{-x}}$$

**Exercice 3** Calculer les limites suivantes :

1  $\lim_{x \rightarrow +\infty} \frac{\ln(1 + e^x)}{x}$

2  $\lim_{x \rightarrow +\infty} \ln(1 + e^{-x})$

3  $\lim_{x \rightarrow +\infty} (\ln(e^x + x) - x)$

4  $\lim_{x \rightarrow 0^+} x \ln x$

5  $\lim_{x \rightarrow 0} \frac{\ln(1 + e^x) - \ln 2}{x}$

6  $\lim_{x \rightarrow +\infty} \frac{\ln x}{e^x}$

**Exercice 4** On considère la fonction

$$f(x) = \ln(e^x + 1) - x.$$

1 Déterminer  $\lim_{x \rightarrow +\infty} f(x)$ .

2 Déterminer  $\lim_{x \rightarrow -\infty} f(x)$ .

3 Interpréter graphiquement ces résultats.